AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous by default. Supervisable by design.

ROBUST USER-FRIENDLY BROAD SEARCH

ROADMAP / TALK ORDER

The run, in four moves

A single route from the problem to the proof.

01

PROBLEM

Why current agents fail

Set the task and its failure modes.

02

SYSTEM

A system for sustained research

Keep experience while resetting the scientist.

03

USER EXPERIENCE

Run the research journey

Move from task contract to retained state.

04

EVALUATION

From claim to evidence

Measure what survives the run.

01 / PROBLEM · THE RESEARCH TASK

Automating ML experimentation

Given a dataset and a scoring metric, an agent must build a model and improve it through repeated experiments.

01	02	03	04	05
Read the problem	**Inspect data**	**Engineer features**	**Train + tune**	**Evaluate**
What should the model predict?	What patterns does the data contain?	Turn raw data into useful model inputs	Fit a model and adjust its settings	Check performance on validation data

REFLECT + REVISE

The agent uses the results to choose what to change, revises the code, and runs the next experiment.

THE AGENT'S JOB

Write the code and run the full loop autonomously.
This includes building the pipeline and deciding which experiments to try next.

THE CHALLENGE GOAL

Reproduce, then improve on the organizer's baseline.
Develop with training data and public validation. The organizer scores the final submission on a hidden test set.

01 / PROBLEM · FAILURE MODES

Why current agents fail

01 WORKFLOW ✓

Observe → Code → **Error** → **Recover**

Coding Harness

Runtime Tools

Acquire missing tools at runtime.

- ✓ Resume after unexpected errors.
- ✓ Extend with Skills & MCP.

02 SINGLE AGENT ✓

last frame —— next move

history-shaped | single path

Fresh reasoning. Audited evidence.

- ✓ Reopen search.
- ✓ Check the science.

03 TREE SEARCH ✓

READ ONCE → BRANCH EDITS

Persist the read. Vary the hypothesis.

Reuse context across experiments.

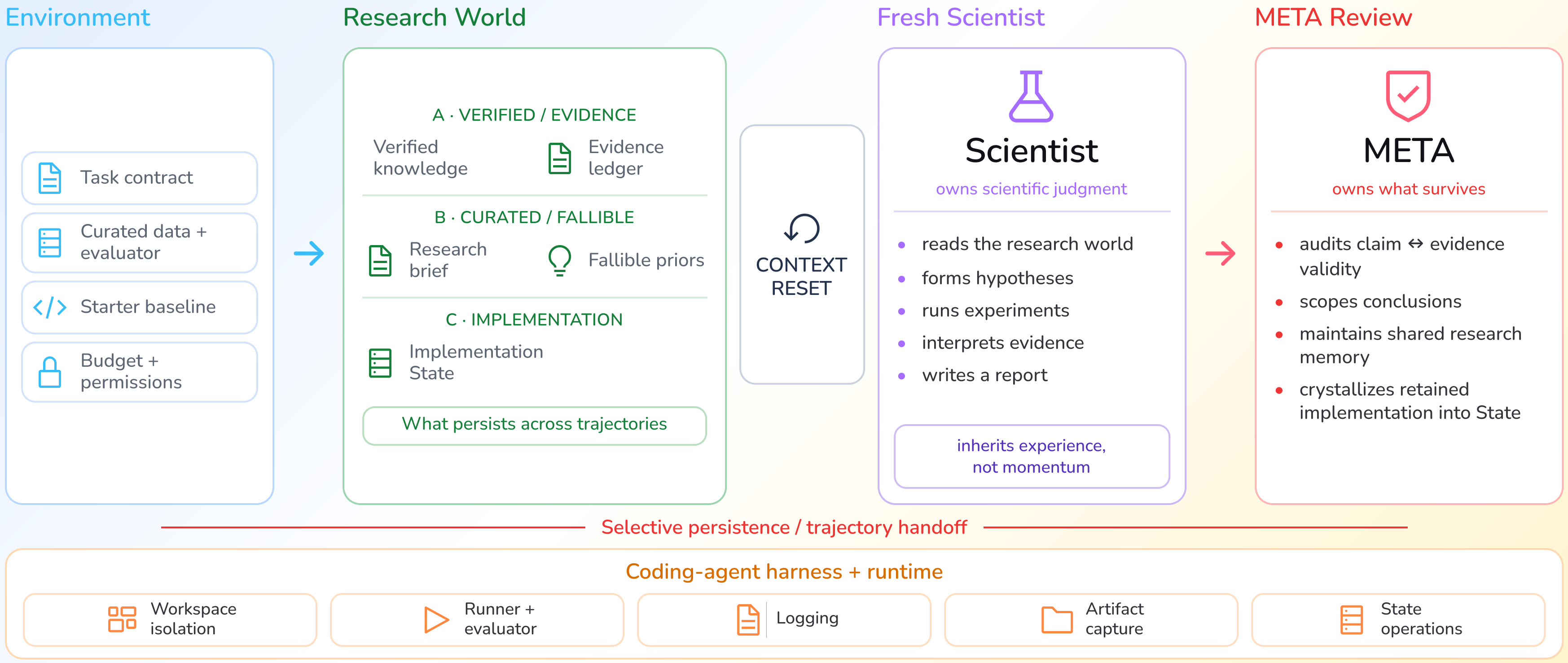
- ✓ Raise task granularity.
- ✓ Parallelize across branches.

- Adaptive tools
- Fresh reasoning
- Audited evidence
- Reusable context

Build the system →

02 / SYSTEM · DESIGN

A system for sustained research



SciOdyssey

A research layer over your agent harness.

BACK TO HAND ×

We are not tokenmaxxing

PROJECT-LEVEL RESOURCE ACCOUNTING

~$10

subscription-equivalent usage from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

COMPUTE

1 × MacBook M2

Development and experiments ran on one consumer laptop.

Retained-run telemetry

48.240M

total tokens

including cache-read input

4.020M

non-cache

input + output

0

GPU-hours

CPU / NumPy training

05 / INSIGHTS · WHAT THE RUN TAUGHT US

The run taught us more than the final score.

Two ways to widen the search without losing evidence.

01

Different agents search differently.

Recorded runs took different routes: local refinement, mechanism pivots, and broader search. Heterogeneous Scientists turn those search styles into a diversity prior. Diversity is a research resource, not a guarantee.

02

Parallel search + review can produce a stronger candidate.

Independent Scientists explore without a shared live context. Review selects a branch, then a fresh Scientist synthesizes the evidence. One observed run moved from ≈0.6015 to 0.6055536 Primary.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS

Give every agent a branch.

main
clean trunk

- agent A / branch
- agent B / branch
- agent C / branch

review
→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP

Leave the next move in GitHub.

01 Issue / prompt — inject the task
→
02 GitHub Action — run the agent
→
03 Commit — review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

PROPOSED PRACTICE Workflow guidance; GitHub continuation is not a benchmarked system feature.

Research is an open-world task.
We need an open-world coding agent.

01 OPEN ACTION SPACE

- Shell
- Files
- Browser
- Git

Use what exists

02 SELF-RECOVERY

- Read errors
- Inspect docs
- Inspect source
- Install packages

Fix what breaks

03 RUNTIME EXTENSION

- MCP
- Skills
- CLI
- Packages

Add what's missing

APPENDIX / THE TEAM

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD

CZ

Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture · integration

ZZ

Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles · evidence review

SX

Shilin Xu

DATA / METRICS

Data workflow · contracts · alignment

GG

GE GAO

RUNTIME / RELIABILITY

Execution · testing · integration support

JL

Jiran Li

EVIDENCE / REPRODUCIBILITY

Telemetry · documentation · packaging

Every handoff kept the research world moving.